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Project Domain	Salesforce, Tableau, GenAI, Power BI & AI/ML	Assessment Standard	Enterprise Benchmarking & UAT

PROJECT TECHNICAL REPORT: PERFORMANCE TESTING & ENTERPRISE BENCHMARKING

Multi-Stack Evaluation: Salesforce Bulk API, Tableau & Power BI Analytics, GenAI Models & UAT Sign-Off

1. Executive Summary & Multi-Stack Environment

This technical evaluation assesses data throughput, transformation latency, model inference response times, and dashboard render stability across interconnected enterprise platforms including Salesforce, Generative AI endpoints, Power BI, Tableau, and Machine Learning pipelines.

- **Salesforce Cloud:** Bulk API v2.0 ingestion into Account, Contact, and Opportunity objects with custom Apex triggers.
- **GenAI / LLM Stack:** Generative AI inference pipeline evaluated for token velocity, time-to-first-token (TTFT), and context window scaling under concurrent prompt requests.
- **Power BI & Tableau Server:** DirectQuery and Hyper In-Memory extract refreshes supporting executive analytics, Level of Detail (LOD) calculations, and complex DAX visual queries.
- **AI / Machine Learning Pipelines:** Batch feature store extraction and distributed Spark model inference engines for predictive business scoring.
- **User Acceptance Testing (UAT):** End-to-end business process verification ensuring complete functional alignment across operational departments.

2. Workload Profiles & Test Scenarios

Scenario ID	Target		Technol-		Workload		Vol-		Execution Mode		Primary SLA Met- ric
	ogy				ume (N)						
TS-SF-01	Salesforce Import	Data			50,000 records				Bulk API v2.0 Parallel		Ingestion Speed $\geq$ 150 rec/sec
TS-AI-02	GenAI Performance	Perfor-			1,000 queries	prompt			Concurrent LLM Inference		Token Velocity $\geq$ 45 tokens/sec
TS-BI-03	Power BI Refresh				5,000,000 rows				Incremental DirectQuery		Refresh Time $\leq$ 120 seconds
TS-TB-04	Tableau Analytics				2,500,000 rows				Hyper Extract Engine		Visual Render $\leq$ 2.5 seconds
TS-ML-05	ML Pipeline				100,000 feature vectors				Distributed Spark ML		Batch Duration $\leq$ 180 seconds
TS-UAT-06	User Acceptance (UAT)	Acceptance			150 Flow	End-to-End			Functional Verification		Pass Rate = 100% Critical Flows

3. Empirical Performance Benchmarking Results

Test ID	Platform / Stack		Duration (t)	Throughput (T)	Avg Latency	Status
TS-SF-01	Salesforce	Bulk API	210.0 sec	238.1 rec/sec	4.20 ms/rec	PASSED
TS-AI-02	GenAI	Inference API	18.2 sec/query	52.4 tokens/sec	19.08 ms/token	OPTIMIZED
TS-BI-03	Power BI Dataset		84.5 sec	59,171 rows/sec	0.017 ms/row	PASSED
TS-TB-04	Tableau Hyper	Extract	1.85 sec (Render)	1,351,351 rows/sec	0.74 ms/view	PASSED
TS-ML-05	ML Batch Pipeline		112.0 sec	892.8 vectors/sec	1.12 ms/vector	PASSED

4. Mathematical Formulas & Throughput Calculations

1. Record / Data Throughput Rate (T)

Quantifies the rate of record ingestion or feature vectorization per unit of time:

$$T = \frac{N}{t} = \frac{50,000 \text{ records}}{210.0 \text{ seconds}} \approx 238.10 \text{ records/second}$$

2. GenAI Token Generation Throughput (T_{token})

Measures the streaming throughput of Large Language Model output generation across concurrent client sessions:

$$T_{\text{token}} = \frac{\sum_{i=1}^k \text{Tokens Output}_i}{t_{\text{generation}}} = \frac{953.68 \text{ tokens}}{18.2 \text{ seconds}} \approx 52.40 \text{ tokens/second}$$

3. Optimization Efficiency Gain (η)

Evaluates processing efficiency improvements achieved through index creation, query refactoring, and caching mechanisms:

$$\eta = \left(\frac{t_{\text{unoptimized}} - t_{\text{optimized}}}{t_{\text{unoptimized}}} \right) \times 100\% = \left(\frac{700\text{s} - 210\text{s}}{700\text{s}} \right) \times 100\% = 70.0\%$$

5. Bottleneck Diagnostics & System Triage

Platform	Identified Bottleneck & Cause	Implemented Optimization
Salesforce Cloud	Governor limits hit during Apex triggers on bulk updates.	Bypassed non-critical triggers; switched to Bulk API v2.0 parallel ingestion.
GenAI / LLM	High TTFT caused by large context window parsing.	Implemented KV-cache reuse, prompt compression, dynamic GPU batching.
Power BI	Slow DirectQuery visuals due to unindexed RLS rules.	Refactored DAX measures, added composite models, materialized summary tables.
Tableau Server	Latency on complex Level of Detail (LOD) expressions.	Pre-computed LODs in Tableau Hyper extracts; scheduled background pre-warming.
Machine Learning	I/O throttling during high-dimensional feature retrieval.	Migrated active feature store tables to in-memory Redis cluster with async fetch.

6. User Acceptance Testing (UAT) Sign-Off Matrix

UAT Module	Acceptance Criteria	Stakeholder Feedback	Sign-Off Status
Salesforce Sync	Bulk records match legacy sources with zero orphan records.	Verified and approved by CRM Ops.	PASSED / SIGNED
GenAI Functional	Output complies with corporate safety & formatting rules.	Approved by Governance Team.	PASSED / SIGNED
Power BI / Tableau	Visual filtering renders within 2.5s with exact fidelity.	Key KPI metrics verified by Executives.	PASSED / SIGNED
AI / ML Scoring	Prediction pipeline delivers precision & recall > 92%.	Validated by Data Science Team.	PASSED / SIGNED

Final Conclusion & Sign-Off Recommendation:

The performance benchmarking and User Acceptance Testing (UAT) across Salesforce, GenAI, Power BI, Tableau, and Machine Learning pipelines confirm that all evaluated modules meet enterprise reliability and SLA criteria. Achieving an overall processing efficiency gain of $\eta = 70.0\%$, the integrated system is officially approved for production deployment.